

study of 13,369 children ages 8-12 living in 24 communities in U.S. and Canada. Health data were collected in 1988-1991, and single-pollutant models were used in the analysis to test a number of measures of particulate air pollution. Dockery et al. found that annual level of sulfates and particle acidity were significantly related to bronchitis, and $PM_{2.1}$ and PM_{10} were marginally significantly related to bronchitis. The original study measured $PM_{2.1}$, however when using the study's results we use $PM_{2.5}$. This makes only a negligible difference, assuming that the adverse effects of $PM_{2.1}$ and $PM_{2.5}$ are comparable. They also found nitrates were linked to asthma, and sulfates linked to chronic phlegm. It is important to note that the study examined annual pollution exposures, and the authors did not rule out that acute (daily) exposures could be related to asthma attacks and other acute episodes. Earlier work, by Dockery et al. (1989), based on six U.S. cities, found acute bronchitis and chronic cough significantly related to PM_{15} . Because it is based on a larger sample, the Dockery et al. (1996) study is the better study to develop a C-R function linking $PM_{2.5}$ with bronchitis.

Bronchitis was counted in the study only if there were "reports of symptoms in the past 12 months" (Dockery et al., 1996, p. 501). It is unclear, however, if the cases of bronchitis are acute and temporary, or if the bronchitis is a chronic condition. Dockery et al. found no relationship between PM and chronic cough and chronic phlegm, which are important indicators of chronic bronchitis. For this analysis, we assumed that the C-R function based on Dockery et al. is measuring acute bronchitis. The C-R function is based on results of the single pollutant model reported in Table 1.

Acute Bronchitis

The estimated logistic coefficient and standard error are based on the odds ratio (1.50) and 95% confidence interval (0.91-2.47) associated with being in the most polluted city ($PM_{2.1} = 20.7 \mu\text{g}/\text{m}^3$) versus the least polluted city ($PM_{2.1} = 5.8 \mu\text{g}/\text{m}^3$) (Dockery et al., 1996, Tables 1 and 4). The original study used $PM_{2.1}$, however, we use the $PM_{2.1}$ coefficient and apply it to $PM_{2.5}$ data.

Incidence Rate: annual bronchitis incidence rate per person = 0.043 (American Lung Association, 2002a, Table 11)

Population: population of ages 8-12.

G.1.5.2 Ostro (1987)

See full study description under Ostro (1987) in Appendix E, Section E.5.1.

G.1.5.3 Ostro and Rothschild (1989)

See full study description under Ostro and Rothschild (1989) in Appendix E, Section E.5.2.

G.1.5.4 Schwartz and Neas (2000)

Schwartz et al. (2000) replicated a previous analysis (Schwartz et al., 1994) linking PM levels to lower respiratory symptoms in children in six cities in the U.S. The original

study enrolled 1,844 children into a year-long study that was conducted in different years (1984 to 1988) in six cities. The students were in grades two through five at the time of enrollment in 1984. By the completion of the final study, the cohort would then be in the eighth grade (ages 13-14); this suggests an age range of 7 to 14. The previous study focused on PM₁₀, acid aerosols, and gaseous pollutants, although single-pollutant PM_{2.5} results were reported. Schwartz et al. (2000) focused more on the associations between PM_{2.5} and PM_{10-2.5} and lower respiratory symptoms. In single and co-pollutant models, PM_{2.5} was significantly associated with lower respiratory symptoms, while PM_{10-2.5} was not. PM_{10-2.5} exhibited a stronger association with cough than did PM_{2.5}. The PM_{2.5} C-R functions for lower respiratory symptoms are based on the results of the reported single pollutant and co-pollutant model (PM_{2.5} and PM_{10-2.5}).

Lower Respiratory Symptoms

The coefficient and standard error are calculated from the reported odds ratio (1.33) and 95% confidence interval (1.11-1.58) associated with a 15 µg/m³ change in PM_{2.5} (Schwartz and Neas, 2000, Table 2).

Incidence Rate: daily lower respiratory symptom incidence rate per person = 0.0012 (Schwartz et al., 1994, Table 2)

Population: population of ages 7 to 14

G.1.6 Asthma-Related Effects

Table G-10 summarizes the additional health impacts functions used to estimate the relationship between PM_{2.5} and asthma exacerbation. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-10. Additional Health Impact Functions for Particulate Matter and Asthma-Related Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Cough	Mar et al.	2004	Spokane, WA	6-18		D24HourMean	0.0191	0.0098	Logistic	Uses incidence rate from Ostro et al. (2001). Age range adjusted.
Shortness of Breath	Mar et al.	2004	Spokane, WA	6-18		D24HourMean	0.0122	0.0138	Logistic	Uses incidence rate from Ostro et al. (2001). Age range adjusted.
Cough	Ostro et al.	2001	Los Angeles, CA	6-18		D24HourMean	0.000985	0.000747	Logistic	
Shortness of Breath	Ostro et al.	2001	Los Angeles, CA	6-18		D24HourMean	0.002565	0.001335	Logistic	
Wheeze	Ostro et al.	2001	Los Angeles, CA	6-18		D24HourMean	0.001942	0.000803	Logistic	
Upper Respiratory Symptoms	Pope et al.	1991	Utah Valley	9-11		D24HourMean	0.003600	0.001500	Logistic	

G.1.6.1 Mar et al. (2004)

Mar et al. (2004) studied the effects of various size fractions of particulate matter on respiratory symptoms of adults and children with asthma, monitored over many months. The study was conducted in Spokane, Washington, a semiarid city with diverse sources of particulate matter. Data on respiratory symptoms and medication use were recorded daily by the study's subjects, while air pollution data was collected by the local air agency and Washington State University. Subjects in the study consisted of 16 adults—the majority of whom participated for over a year—and nine children, all of whom were studied for over eight months. Among the children, the authors found a strong association between cough symptoms and several metrics of particulate matter, including PM_{2.5}. However, the authors found no association between respiratory symptoms and PM of any metric in adults. Mar et al. therefore concluded that the discrepancy in results between children and adults was due either to the way in which air quality was monitored, or a greater sensitivity of children than adults to increased levels of PM air pollution.

Asthma Exacerbation, Cough

In a single-pollutant model, the coefficient and standard error are estimated from the odds ratio (1.21) and 95% confidence interval (1.00-1.47) for a 10.0 µg/m³ increase in 1-day lagged concentration of PM_{2.5} (Mar et al., 2004, Table 7).

Incidence Rate: Daily cough rate per person = 14.5%. Mar et al. (2004) did not report the incidence rate for each type of asthma exacerbation. The daily cough rate from Ostro et al. (2001, p.202) is applied here.

Population: The study reported results for population ages 7-12. For comparability to other studies, we apply the results to the population of ages 6 to 18. We treat this as two groups based on the available information from American Lung Association (2010b, Table 7). Asthmatic population ages 6 to 17 = 10.70% of population ages 6 to 17 and asthmatic population age 18 = 7.19% of population age 18. The American Lung Association (2010b, Table 7) estimates asthma prevalence for children 5- 17 and adults 18-44 at 10.70% and 7.19% respectively (based on data from the 2008 National Health Interview Survey).

Asthma Exacerbation, Shortness of Breath

In a single-pollutant model, the coefficient and standard error are estimated from the odds ratio (1.13) and 95% confidence interval (0.86-1.48) for a 10.0 µg/m³ increase in current-day concentration of PM_{2.5} (Mar et al., 2004, Table 7).

Incidence Rate: Daily shortness of breath rate per person = 7.4%. Mar et al. (2004) did not report the incidence rate for each type of asthma exacerbation. The daily rate of shortness of breath from Ostro et al. (2001, p.202) is applied here.

Population: See the population description for “Asthma Exacerbation, Cough” from Mar et al. (2004).

G.1.6.2 Ostro et al. (2001)

Ostro et al. (2001) studied the relation between air pollution in Los Angeles and asthma exacerbation in African-American children (8 to 13 years old) from August to November 1993. They used air quality data for PM₁₀, PM_{2.5}, NO₂, and O₃ in a logistic regression model with control for age, income, time trends, and temperature-related weather effects. The authors note that there were 26 days in which PM_{2.5} concentrations were reported higher than PM₁₀ concentrations. The majority of results the authors reported were based on the full dataset. These results were used for the basis for the C-R functions. Asthma symptom endpoints were defined in two ways: “probability of a day with symptoms” and “onset of symptom episodes”. New onset of a symptom episode was defined as a day with symptoms followed by a symptom-free day.

The authors found cough prevalence associated with PM₁₀ and PM_{2.5} and cough incidence associated with PM_{2.5}, PM₁₀, and NO₂. Ozone was not significantly associated with cough among asthmatics. The authors found that both the prevalent and incident episodes of shortness of breath were associated with PM_{2.5} and PM₁₀. Neither ozone nor NO₂ were significantly associated with shortness of breath among asthmatics. The authors found both the prevalence and incidence of wheeze associated with PM_{2.5}, PM₁₀, and NO₂. Ozone was not significantly associated with wheeze among asthmatics.

The derived health impact functions are based on the results of single pollutant models looking at the probability of symptoms.

Asthma Exacerbation, Cough

The coefficient and standard error are based on an odds ratio of 1.03 (95% CI 0.98-1.07) for a 30 µg/m³ increase in 12-hour average PM_{2.5} concentration (Ostro et al., 2001, Table 4, p.204).

Incidence Rate: daily cough rate per person (Ostro et al., 2001, p.202) = 0.145

Population: The study reported results for African-American population ages 8-13. For comparability to other studies, we apply the results to the African-American population of ages 6 to 18. We treat this as two groups based on the available information from American Lung Association (2010b, Table 9). Asthmatic African-American population ages 6 to 17 = 17.76% of African-American population ages 6 to 17 and asthmatic African-American population age 18 = 7.52% of African-American population age 18. The American Lung Association (2010b, Table 9) estimates asthma prevalence for children 5- 17 and adults 18-44 at 17.76% and 7.52% respectively (based on data from the 2008 National Health Interview Survey).